Truth as a Compression Artifact
in Language Model Training

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Models prefer the correct answer *only* when wrong answers cannot be compressed.

Truth bias is not an inherent property of next-token prediction — it tracks the description length of the false answer system.

Random errors
65% → 85%
truth bias *scales* with model size

One coherent rule
∼45–51%
truth bias *vanishes* at all scales

Two coherent rules
47% → 78%
truth bias *returns* – selector cost

1. The Question

Real-world corpora contain contradictions: the same problem appears with both correct and incorrect solutions. Yet models trained on such data often *prefer* the correct one. **Why?**

Existing answers (scaling, frequency, truth-correlated representations) do not explain why the **training objective alone** would favor one answer over another at *equal frequency in the same format*.

2. Hypothesis – MDL Framing

Cross-entropy minimization ≡ minimum code length
(Shannon 1948; Deletang et al. 2024).

An MDL learner minimizes

$$L(M) + L(D \mid M)$$

In our task families:

- **Random errors:** $K(T_2) \gg K(T_1)$ – false cluster needs $O(n)$ bits ⇒ truth wins.
- **Coherent errors:** $K(T_2) \approx K(T_1)$ – both equally compressible ⇒ no preference
- ***N* rules + selector:** adds $O(n \log N)$ bits ⇒ preference returns.

The compression objective tracks **consistency**, not truth.

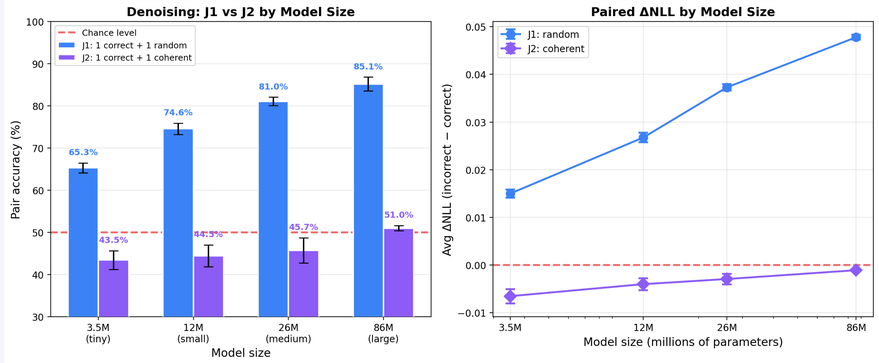
3. Experimental Design

GPT-2-like decoder-only transformers, trained *from scratch*. Each problem appears with *both* a correct and an incorrect solution; only the *structure of errors* varies.

Config	Layers	d_{model}	Params
tiny	4	256	∼3.5 M
small	6	384	∼12 M
medium	8	512	∼26 M
large	12	768	∼86 M
Qwen3	28	–	0.42 B / 1 B

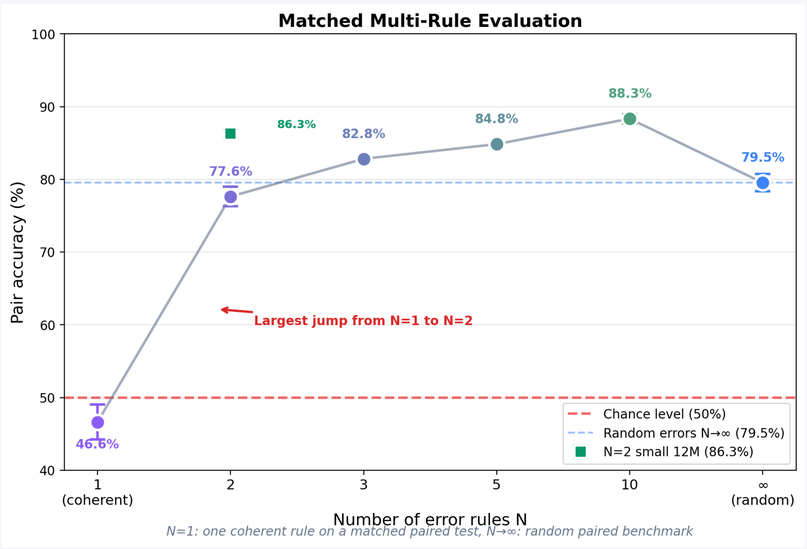
AdamW, lr = $3 \cdot 10^{-4}$, 5,000 steps, 4 seeds per condition. **Metric:** paired accuracy – fraction of pairs where the model assigns lower NLL to the correct completion (length-matched).

4. Central Contrast: Random vs. Coherent



Random errors: accuracy scales monotonically: 65.3%→74.6%→81.1%→85.2% (3.5–86 M). **Coherent** errors: accuracy stays at chance across all sizes (43.5%→51.0%). **A single coherent rule erases the preference.**

5. Multi-Rule Crossover (the smoking gun)



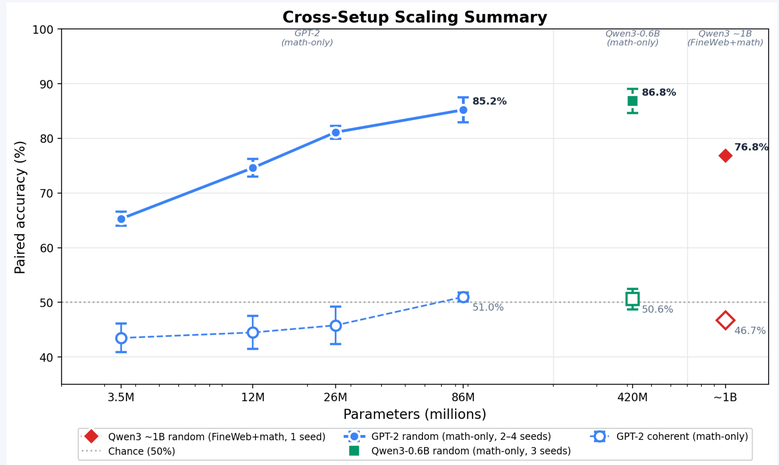
Sharp transition at the compressibility boundary: $N=1$ gives chance, $N=2$ jumps to 77.6%, $N=10$ reaches 88.3% – *above* pure random (79.5%). Each rule is compact, but the *selector* (which rule applies to which problem) costs $\log N$ bits per problem – exactly the MDL prediction. **Hard to reconcile** with lexical-diversity or token-corruption explanations.

6. Transfer: Wikipedia Entity Substitution

Size	Random	Coherent
tiny (3.5 M)	**69.6 %**	48.7 %
small (12 M)	**70.7 %**	46.6 %
medium (26 M)	**71.5 %**	46.4 %
large (86 M)	**71.4 %**	45.9 %

Same contrast on natural text: random substitution of named entities yields ∼71 %; a *coherent* global mapping ("France"→"Japan", "Paris"→"Tokyo") yields chance.
The compressibility gap is smaller in language than in math (+22 pp vs. +34 pp) – locally fluent errors compress better.

7. Scaling to 1 B Parameters on Real Text



Effect persists at ∼1 B parameters on a mixed FineWeb-Edu + math corpus (math = 8 % of training): **76.8 %** random vs. **46.7 %** coherent. Pattern reproduces across architectures (GPT-2, Qwen3) and across synthetic vs. realistic mixtures.

8. Takeaway

why this matters for alignment, hallucinations, and data curation

- **Truth bias is not intrinsic to next-token prediction.** It is a **compression artifact**: it appears precisely when false answers are *harder to compress* than true ones.
- **Coherent misinformation** – internally consistent, just wrong – evades the filter at *equal frequency*.
- **Verification helps but *inversely scales***: bigger models absorb coherent within-domain patterns more readily, making post-hoc fixes *less* effective.
- **Practical implication:** diverse errors filter out, *coordinated* errors may not. A relevant signal for alignment, data curation, and hallucination control at scale.